# Irregular polygonal Radon planes with every admissible number of vertices

Candidate full solution for arXiv:1809.04760

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**Status:** full solution likely valid, subject to human review.

## 1 Source question

Mandal, Sain, and Paul prove that every Euclidean regular polygon with $4n+2$ vertices is the unit sphere of a Radon plane and construct irregular hexagonal examples. They then ask the following question on page 11 of arXiv:1809.04760v2 [1].

> Given any $n \in \mathbb{N}$, does there exist a Radon plane $X$ such that $S_X$ is an irregular polygon with $4n+2$ vertices? If the answer is affirmative, find a generalized algorithm constructing such planes.

Here "regular" has the source's Euclidean meaning: all side lengths and all interior angles agree. We take $\mathbb{N} = \{1, 2, \ldots\}$, as in the source's polygonal statements.

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**Remark 2.2.** If the vertices of $S_{\mathbb{X}}$ in the above theorem are $\pm(1,\alpha), \pm(1,-\alpha)$ and $\pm(0,2\alpha)$ then $\mathbb{X}$ is a Radon plane with regular polygonal unit ball if and only if $\alpha = \frac{1}{\sqrt{3}}$. On the other hand, if the vertices of $S_{\mathbb{X}}$ are $\pm(1,\alpha), \pm(1,-\alpha)$ and $\pm(2,0)$ then $\mathbb{X}$ is a Radon plane with regular polygonal unit ball if and only if $\alpha = \sqrt{3}$. In all other cases, $\mathbb{X}$ is a Radon plane whose unit ball is an irregular polygon.

**Conclusion:** In this paper we have obtained a complete geometric characterization of polygonal Radon planes. Furthermore, we have given concrete examples of Radon planes for which the unit spheres are polygons, but not regular polygons. Motivated by these findings, we end the present paper with the following:

**Open question:** *Given any $n \in \mathbb{N}$, does there exist a Radon plane $\mathbb{X}$ such that $S_{\mathbb{X}}$ is an irregular polygon with $4n+2$ vertices? If the answer to this question is in the affirmative, then find a generalized algorithm to construct Radon planes whose unit spheres are irregular polygons with $4n+2$ vertices.*

Figure 1: The open question in the original paper, page 11.

## 2 Proof intuition

The Radon property belongs to the normed linear structure, so it is invariant under linear isometry. Euclidean regularity of a drawn unit polygon is not: an anisotropic stretch changes its side lengths.

Start with the regular $(4n+2)$-gon, which the source already proves is Radon, and transport its norm by the diagonal map $(x, y) \mapsto (\lambda x, y)$. The transported plane is isometric to the original Radon plane, but two explicitly chosen edges of its unit polygon have unequal Euclidean lengths whenever $\lambda \neq 1$.

## 3 Formal result

Recall that $x \perp_B y$ in a normed space if $\|x + ty\| \geq \|x\|$ for every $t \in \mathbb{R}$. A real normed plane is a Radon plane when this relation is symmetric.

**Lemma 1** (Transport of Birkhoff–James orthogonality)**.** *Let $(V, \|\cdot\|)$ be a real normed space, let $T : V \to V$ be invertible and define $\|z\|_T = \|T^{-1}z\|$. Then*

$$x \perp_B y \quad \Longleftrightarrow \quad Tx \perp_B Ty$$

*where the relation on the right is computed using $\|\cdot\|_T$. In particular, the Radon property is preserved.*

*Proof.* For every real $t$,

$$\|Tx + tTy\|_T = \|T^{-1}(Tx + tTy)\| = \|x + ty\|, \qquad \|Tx\|_T = \|x\|.$$

Thus the defining inequalities for the two orthogonality statements are identical. Symmetry is therefore transported in both directions. □

**Theorem 1** (Uniform affirmative answer)**.** *For every integer $n \geq 1$ and every $\lambda > 0$ with $\lambda \neq 1$, there is a Radon norm on $\mathbb{R}^2$ whose unit sphere is an irregular polygon with exactly $4n+2$ vertices. More precisely, with $m = 4n + 2$, let*

$$v_k = \left(\cos \frac{2\pi k}{m}, \sin \frac{2\pi k}{m}\right), \quad 0 \leq k < m,$$

*let $P_m = \operatorname{conv}\{v_0, \ldots, v_{m-1}\}$, and set*

$$T_\lambda(x, y) = (\lambda x, y), \qquad B_{n,\lambda} = T_\lambda(P_m).$$

*Then $B_{n,\lambda}$ is the unit ball of such a norm.*

*Proof.* Since $m$ is even, $P_m$ is centrally symmetric about the origin and has the origin in its interior, so its Minkowski functional is a norm. The source paper proves that this regular $m$-gonal normed plane is Radon [1, Theorem 2.7]. The set $B_{n,\lambda}$ is again a centrally symmetric convex body with the origin in its interior and hence is a unit ball. Its gauge is

$$\|z\|_{n,\lambda} = \|T_\lambda^{-1}z\|_{P_m}.$$

Consequently $T_\lambda$ is a linear isometry from the regular polygonal norm onto this new norm. The lemma shows that the new plane is Radon. Because $T_\lambda$ is invertible, it maps the vertices and edges of $P_m$ bijectively to those of $B_{n,\lambda}$; hence the latter has exactly $m = 4n + 2$ vertices.

It remains to show that this polygon is not Euclidean regular. Put $a = \pi/m$. The $k$th original edge vector satisfies

$$v_{k+1} - v_k = 2 \sin a \left(-\sin((2k+1)a),\ \cos((2k+1)a)\right),$$

where indices are taken modulo $m$. Its image therefore has Euclidean length

$$L_k = 2\sin a\sqrt{\lambda^2\sin^2((2k+1)a) + \cos^2((2k+1)a)}.$$

For $k = n$, the identity $(2n+1)a = \pi/2$ gives

$$L_n = 2\lambda\sin a,$$

whereas

$$L_0 = 2\sin a\sqrt{\lambda^2\sin^2 a + \cos^2 a}.$$

If $L_0 = L_n$, squaring and cancelling $4\sin^2 a$ would yield

$$(1-\lambda^2)\cos^2 a = 0.$$

But $m \geq 6$, so $0 < a \leq \pi/6$ and $\cos a \neq 0$, while $\lambda \neq 1$. This is impossible. Thus two sides have different lengths, so $B_{n,\lambda}$ is irregular in the source's sense. □

# 4 Construction algorithm

Given $n \geq 1$, set $m = 4n+2$ and output the vertices

$$\boxed{w_k = \left(2\cos\frac{2\pi k}{m},\ \sin\frac{2\pi k}{m}\right), \quad k = 0, \ldots, m-1.}$$

Their convex hull is the unit ball. This is the preceding theorem with the fixed choice $\lambda = 2$, so no search or case distinction is needed.

# 5 Verification notes

The argument is symbolic and has no numerical dependency. The accompanying regression script is stored at

`code/check_edge_lengths.py`.

It evaluates all image edge lengths for $n = 1, \ldots, 100$ at $\lambda = 2$ and checks both the displayed formula and $L_0 \neq L_n$. It is a regression test only, not part of the proof.

The main human-review points are: (i) the source theorem for the regular $(4n+2)$-gon; (ii) invariance under the transported norm; and (iii) that the source uses "irregular" in the Euclidean, not affine, sense. No stronger claim about classification of all irregular polygonal Radon planes is made.

# 6 Bounded novelty check

The search, performed through 11 August 2026, covered the run's registry, solution, attempt, and proof-gap indexes; exact-title and exact-question web searches; arXiv queries for polygonal Radon planes; and combinations of the phrases *irregular*, $4n+2$, *affine image*, and *affinely regular*. No later paper explicitly answering the exact all-orders question was found. Search results do recognize affine-regular hexagonal Radon spheres, and the isometric-invariance observation is elementary. Accordingly, the result is a complete answer to the stated question, but its novelty confidence is only moderate-to-low pending a specialist citation check.

## 7 Conclusion and review recommendation

The source question has an affirmative answer for every $n \geq 1$, with the uniform deterministic construction $T_2(P_{4n+2})$. We recommend promotion as `full_solution_likely_valid`, subject to human verification of the terminology and the bounded novelty search.

## References

[1] K. Mandal, D. Sain, and K. Paul, *A complete characterization of Radon planes whose unit spheres are regular polygons*, arXiv:1809.04760v2, 2018. https://arxiv.org/abs/1809.04760.